Edge-Deployable LLMs for *Natural Language → SBOL3 Design*

"A fine-tuned 27B model generating wet-lab-ready genetic circuit designs, running locally on a 15-watt Jetson Orin NX."

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EC/BE 552 — COMPUTATIONAL SYNTHETIC BIOLOGY

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Turning plain English into *lab-ready DNA designs.*

THE PROBLEM

- Draw genetic circuit on GUI
- Frontier LLMs only live in the cloud
- Biosafety-sensitive lab, an undergrad without a GPU budget, or a teaching environment with network restrictions

THE GOAL

A natural-language → SBOL3 pipeline that is:

Local — runs offline on commodity / edge hardware

Standard Output — every output parses as valid SBOL 3.1 RDF/XML

Benchmarked — 100-prompt, 6-axis deterministic rubric

Honest — structural quality vs. biological quality reported separately

REAL-LIFE USE CASE

"A HEK293 synNotch receptor that detects **CD19 on tumor cells**, *drives* **IL-12 expression**, *with a* **rapamycin kill-switch**."

45 seconds later → valid SBOL3 RDF/XML → piped directly into her existing assembly software.

Patient data never leaves the building. No API key. No cloud. No $0.07-per-circuit meter.

WHY SYNTHETIC BIOLOGY NEEDS THIS

- Circuit design & Brainstorm
- Cutting design time from days to seconds
- Offline, at zero marginal cost

90% *"of frontier performance on a $1,200 Jetson Orin NX —* ***offline, 15 watts sustained, $0 per circuit after hardware.****"*

What it does, *in one breath.*

> *"You describe a genetic circuit in English. Newgenes returns a **validated SBOL 3.1 RDF/XML file** that any SBOL-compatible tool can import — generated **entirely on a 15-watt edge device.**"*

STEP 01 · PROMPT

Prompt

"Build a LacI/TetR/cI repressilator oscillator in E. coli."

STEP 02 · MODEL

Model

MAC MLX
Qwen 3.5 27B + LoRA · Q8

JETSON ORIN NX
Gemma-4 26B-A4B + LoRA · Q3

STEP 03 · SBOL3

SBOL3

Valid RDF/XML 16
SubComponents · 12
Interactions

What it does, *in numbers.*

PERFORMANCE SPEC · TARGET VS. ACHIEVED

METRIC	TARGET	ACHIEVED
Avg time / circuit	< 2 min	45 s Mac · 96 s Jetson
Training cost	< $300	$260 (OpenAI API, data gen)
Inference power	≤ 15 W	15 W sustained (Jetson)
Inference cost	$0 marginal	$0 after hardware
Hardware floor	< $1,500	$1,200 Jetson Orin NX
Quality floor	≥ 75 / 100	78 / 100 worst case (LoRA)
Quality mean	≥ 85 / 100	89.6 Jetson · 91.6 Mac
Internet required	none	none — fully offline

APPLICATION REQUIREMENTS

- **Python 3.10+**
- `mlx_lm` (Mac) or `llama.cpp` (Jetson) — local server on :8080
- ~20 GB disk for quantized model + LoRA adapter
- **Zero internet** at inference time

CONSTRAINTS WE LIVED INSIDE

- **16 GB RAM** (Jetson) / **64 GB** (Mac) → forced Q3 / Q8 quantization
- No cloud GPU budget → LoRA on a consumer laptop, not A100 full fine-tune
- One-family scope — budget only allowed end-to-end test of one model line
- Deterministic eval required — hand-coded 6-axis rubric, no LLM-as-judge in the headline

INTERFACE CONTRACT

INPUT	English prompt + canonical system prompt (~1,120 chars)
OUTPUT	SBOL 3.1 RDF/XML + JSON
SCHEMA	`{ name, organism, components[], interactions[], behavior }`
GUARANTEES	valid JSON · unique IDs · referential integrity · SBOL 3.1 schema-pass · all URIs resolve
ON FAIL	one schema-retry with error context. No second retry.
DETERMINISM	LLM step probabilistic · **JSON → SBOL3 fully deterministic**

Training data — *1,162 examples, three provenance tiers.*

Goal of this step: *teach the model that "SBOL JSON" is a specific structured format — not a free-form paragraph.*

TIERS BY EXACT PROMPT OVERLAP

Splits · 1,162 TRAIN / 64 VALID / 75 TEST

SYNTH-SIMPLE · 712 · 61% | SYNTH-COMPLEX · 326 · 28% | SCRAPED · 124 · 11%

TIER 1 · SYNTH-SIMPLE

712

Qwen 2.5 72B, local — $0

Easy circuit prompts (toggle switches, single-input sensors) paired with hand-templated JSON answers.

Fast, covers the easy case cheaply.

TIER 2 · SYNTH-COMPLEX

326

GPT-5.4 API — paid

Harder multi-gate cascades, feedback loops, multi-cellular designs — with rationale text attached.

Catches the long-tail prompts.

TIER 3 · SCRAPED REAL

124

SynBioHub · iGEM · DataCurationProject

Authentic SBOL pulled, then back-translated into natural-language prompts.

Gold examples — real designs real scientists published.

FORMAT

All three tiers → chat-style {system, user, assistant} JSONL.

CONTAMINATION CHECK

100 eval prompts run against every training row: **zero substring overlap** · **max Jaccard 0.47** . The model never saw the eval set.

LoRA + Chen & Truong essay

*Key insight from our 2×2 ablation: the two levers are **~80% redundant on easy circuits, complementary on the tail.***

On the 2×2: LoRA alone +2.50 · essay alone +2.97 · combined only +3.47, not +5.47. Overlapping patterns on the average — but different failure modes on the tail, so we ship both.

PART 1 · QLORA FINE-TUNING

A ~384 MB adapter, trained on a laptop

Rank 8, α 16, last 16 of 64 transformer layers (attention + MLP)

2,500 iters (~1.76 epochs), batch 1 with grad accumulation

MLX on M5 Max 64 GB — 7.5 h, ~$3 of electricity, no cloud

Output: ~384 MB `adapter.safetensors` fused at deploy

What the LoRA actually learned: *valid JSON structure, SBOL part naming conventions, rare circuit patterns from scraped tier → eliminates the 2/100 catastrophic zeros the bare model produces.*

PART 2 · CHEN & TRUONG 2026 ESSAY PROMPT (ONLY USED FOR MACBOOK DEPLOYMENT DUE TO RAM CONSTRAINT)

~4,780 chars, applied at inference

01 Complete SBOL ontology (promoter / RBS / CDS / terminator / operator / regulator)

02 Explicit JSON schema with required fields (parts, interactions, description)

03 Behavioral wiring rules (how activation / repression / transcription arrows must connect)

04 Host-organism appropriateness (E. coli vs. HEK293 vs. yeast parts)

05 Formatting discipline — *"return JSON only, no markdown fences, no preamble."*

BEFORE — BARE GEMMA, DEFAULT PROMPT

87.17 avg · **2/100 catastrophic** · **30%** wrapped in ```` ```json ```` fences

AFTER — LORA

89.60 avg · **0 catastrophic** · **floor 78** · **100%** clean JSON

*Average moves +2.4. Worst case moves from **0 to 78**. That's the production-readiness difference — tail risk, not the mean.*

End-to-end design & *the 6-axis rubric.*

What runs on the laptop, and how every output is scored — same Python code for every model in the paper.

WHAT ACTUALLY RUNS ON THE LAPTOP

- **English prompt**
  "Design a toggle switch..."
- **Chen & Truong essay as system message**
  ~4,780 chars
- **Qwen 3.5 27B + LoRA · Q8 · MLX**
  or Gemma-4 26B-A4B · Q3 · llama.cpp
- **SBOL JSON**
  parts + interactions + description
- **Deterministic 6-axis rubric**
  sbol_eval_v2.py
- **Score 0–100 + circuit diagram PNG**
  networkx + matplotlib render

SBOL_EVAL_V2 — 6 AXES, DETERMINISTIC, 100 PTS

BYTE-STABLE

AXIS	PTS	WHAT IT CHECKS
Structural Validity	20	JSON parses; every referenced part exists
Behavioral Wiring	20	Graph topology valid; activation / repression arrows terminate correctly
Biology Appropriate	20	Parts match the declared host organism
Part Fidelity	20	Part names resolve against known SBOL vocab
Description Quality	10	Rationale text matches the generated circuit
Response Format	10	Clean JSON, no prose wrapping
Total	100	

HEADLINE RESULT

92.21 / 100 Mac · **89.60 / 100** Jetson · vs. **99.22** Opus ceiling. 100 prompts, unseen at training.

Why deterministic scoring matters: *no LLM-as-judge in the headline number. Every point is verifiable Python — schema validators, graph traversals, set intersections. We can re-run any score and get the same number.*

Two-layer *architecture.*

Edge / efficiency layer (local) and quality-ceiling layer (cloud) — complementary, not competing.

USER PROMPT

EDGE / EFFICIENCY LAYER

Local, offline, $0 variable

Mac MLX
Qwen 3.5 27B + LoRA · Q8 — 17.6 tok/s

Jetson Orin NX
Gemma-4 26B-A4B + LoRA · UD-Q3_K_M — 7 tok/s · 15 W · $1.2k

QUALITY-CEILING LAYER

Cloud API, highest quality

Anthropic API
Claude Opus 4.7 + Chen & Truong prompt-only template (4,780 chars)

~50 tok/s · ~$6.5 / 100 designs

JSON → deterministic `json_to_sbol3.py` → SBOL 3.1 RDF/XML

End-to-end *data flow.*

Every arrow crosses either a file boundary or a model invocation..

01 User prompt
English · one-shot · ≤ 1,024 tokens

02 Canonical system prompt
1,120 chars, matches training distribution exactly

03 Inference
`llama-server --lora · rep_penalty 1.05 · max_tokens 4,000`

04 Schema-retry loop
Validate JSON → retry once with error context on fail

05 JSON intermediate
`{ name, organism, components[], interactions[], behavior }`

06 json_to_sbol3.py
Deterministic · SubComponent, Interaction, URIs, schema-valid RDF/XML

LacI / TetR / cI *repressilator* — full round-trip.

Classic Elowitz 2000 topology. ***16 SubComponents, 12 Interactions, all URIs resolve, SBOL 3.1 schema-valid*** *— zero manual cleanup between prompt and RDF. This is the full pipeline working end-to-end on one laptop.*

> *"Design a three-repressor oscillator (LacI, TetR, cI) in E. coli with GFP as the reporter, using a classic repressilator topology."*

JSON INTERMEDIATE (ABRIDGED)

```
{
  "name": "laci_tetr_ci_circuit",
  "organism": "Escherichia coli",
  "components": [
    { "id": "p_laci", "type": "promoter" },
    { "id": "p_tetr", "type": "promoter" },
    { "id": "p_ci",   "type": "promoter" },
    { "id": "laci",   "type": "cds" },
    { "id": "tetr",   "type": "cds" },
    { "id": "ci",     "type": "cds" }
    // ... 10 more (rbs, terminator, gfp ...)
  ],
  "interactions": [
    { "src": "laci", "dst": "p_tetr", "type": "represses" },
    { "src": "tetr", "dst": "p_ci",   "type": "represses" },
    { "src": "ci",   "dst": "p_laci", "type": "represses" }
    // ... 9 more
  ]
}
```

SBOL 3.1 RDF/XML (ABRIDGED)

```
<sbol3:Component rdf:about=".../laci_tetr_ci_circuit">
  <sbol3:displayId>laci_tetr_ci_circuit</sbol3:displayId>
  <sbol3:type rdf:resource="SBO:0000241"/>
  <sbol3:hasFeature>
    <sbol3:SubComponent rdf:about=".../p_laci">
      <sbol3:displayId>p_laci</sbol3:displayId>
      <sbol3:role rdf:resource="SO:0000167"/>
      <sbol3:instanceOf rdf:resource="..."/>
    </sbol3:SubComponent>
  </sbol3:hasFeature>
  <!— 15 more SubComponents —>
  <sbol3:hasInteraction>
    <sbol3:Interaction>
      <sbol3:type rdf:resource="SBO:0000169"/>
      <!— inhibition, LacI ⊣ pTetR —>
    </sbol3:Interaction>
  </sbol3:hasInteraction>
  <!— 11 more Interactions —>
</sbol3:Component>
```

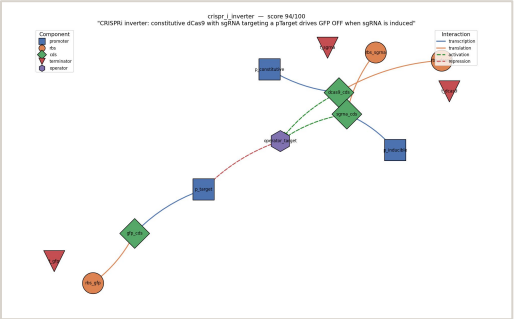
16 SubComponents · 12 Interactions · SBOL 3.1 schema VALID · all URIs resolve

Three hard circuits — *all generated on Jetson.*

All d5 (hardest tier) prompts. JSON parses → directed graph renders automatically. Circuits below rendered straight from model output.

01 · CRISPRI INVERTER

94 / 100

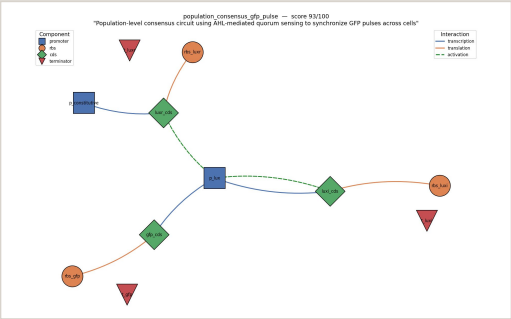


> *"Constitutive dCas9 + sgRNA targeting pTarget drives GFP OFF when sgRNA is induced."*

Two transcription units. GFP on only when sgRNA absent → inverter logic.

02 · QUORUM SENSING

93 / 100

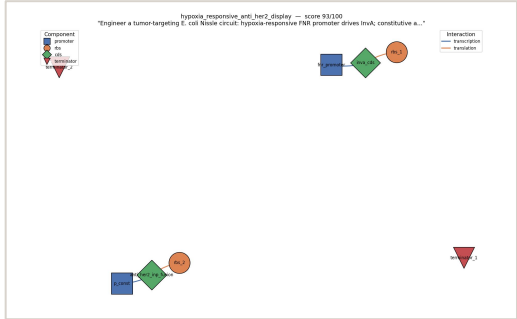


> *"Population-level AHL-mediated consensus — synchronized GFP pulses."*

Positive-feedback loop: luxr → p_lux → luxi. **Bare model: 0/100.**

03 · TUMOR-TARGETING

93 / 100



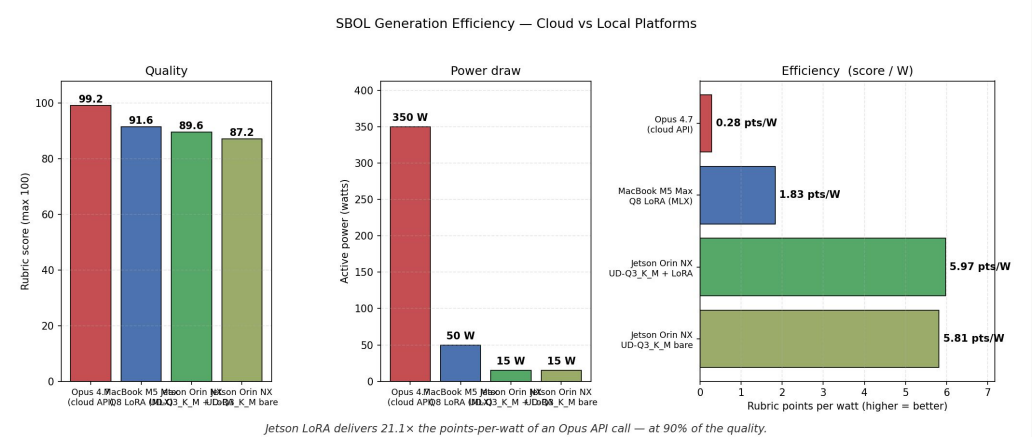
> *"Hypoxia-responsive FNR → InvA; constitutive anti-HER2 nanobody surface-displayed."*

Two independent TUs. Hypoxia branch + targeting branch.

No cleanup, no human edits. *Each JSON parses → render script produces diagram → every topology is biologically legible.*

Quality, power, efficiency — *one chart.*

Same 100-prompt rubric. Same four platforms. Three views of the tradeoff.



Jetson LoRA delivers 21.1× the points-per-watt of an Opus API call — at 90% of the quality.

90%

OF FRONTIER QUALITY

Jetson Gemma-4 Q3 + LoRA scores 89.6 vs. Opus 99.2.

21×

MORE POINTS / WATT

5.97 pts/W on Jetson + LoRA vs. 0.28 pts/W via Opus API.

15 W

ACTIVE POWER BUDGET

Gemma-4 26B-A4B at 7 tok/s on a $1,200 device — fanless, offline.

Evaluation disjoint from training: 0/100 exact overlap · 0/100 substring · max Jaccard 0.47 (topology, not prompt leakage).

What we underestimated — *and what we'd do differently.*

Process lessons from this cycle, and the concrete next iteration.

WHAT WE UNDERESTIMATED

The computational cost of training.

Original plan: *"train overnight."*

Reality: **7.5 h per successful LoRA** + 3–4 failed runs before hitting the right rank / α / layer combo.

Total clock time: **~40 h** across the project.

If we'd had to rent A100s, the **$260 budget would have broken in week 1** .

> *It is reasonable to infer there is still room for further improvement — 1,162 training examples is small, our rubric is structural (not biological), and we have* **not wet-lab validated a single output**.

CONCRETE NEXT ITERATION

01 **Three host-specific LoRAs (bacterial / mammalian / yeast)**

Auto-routed by detected organism. Error analysis shows **60% of remaining failures** are host-inappropriate parts (T7 in mammalian, etc.). ~4 extra training hours — plausibly closes half the gap to Opus.

02 **Add a 7th (logic-fidelity) rubric axis — *during training*, not just eval.**

Direct fix for the NAND-computes-OR failures on kill-switches.

03 **Wet-lab feedback loop.**

Even a small transfection-assay dataset would turn the rubric from structural to biological.

04 **Bigger, more diverse dataset.**

Especially more scraped real-world **mammalian + multicellular** designs.

The 20-point *structural–biology gap.*

Valid JSON is not the same as valid biology — and sometimes training makes it worse.

STRUCTURAL RUBRIC

92.2

on hardest 11 prompts — schema + wiring + syntax

–20.4

EXPERT BIOLOGY RUBRIC

71.4

same outputs, judged by Opus on biological correctness

FAILURE MODES (LOGIC INVERSIONS)

Toggle switches that self-repress
Structurally correct; each CDS incorrectly represses its own promoter.

NAND kill-switches that compute OR
*Wires present, logic inverted. Kill-switch topology **regressed –4 pts** after fine-tuning.*

CRISPR sensors that confuse cleavage
Direct vs. collateral cleavage named inconsistently — 8.7 pt gap to frontier on CRISPR topology.

Measured on Mac Q8 Cell D outputs. Jetson 4B likely shows a comparable or larger gap — proposed 7th rubric axis would apply to both.

LoRA ≈ Chen prompt on easy circuits. *But nfot on hard ones.*

2×2 ablation (n=34, Mac Q8). Then: what happens when we push to Q3 on Jetson.

	DEFAULT PROMPT	CHEN & TRUONG PROMPT
Base (no LoRA)	A 89.71 / 100	B 92.68 / 100
+ LoRA (Q8)	C 92.21 / 100	D 93.18 / 100

EFFECT DECOMPOSITION

LoRA alone (A→C)	+2.50
Chen prompt alone (A→B)	+2.97
Combined (A→D)	+3.47
Sum if additive	*+5.47*
Overlap (redundant gain)	**~2.0 pt**

Chen prompt captures ~85% of LoRA's gain at zero training cost.

Practical rule. No GPU budget? Use the prompt. Have a GPU and need edge deploy? LoRA. Don't stack them expecting additivity.

BUT: TAIL RISK CHANGES THE STORY

At Q8 on Mac, LoRA and prompt are **~80% redundant**.

At **Q3 on Jetson**, bare has two catastrophic parse failures (scores 0). LoRA has zero failures.

LoRA is not redundant where it matters most: hard circuits under aggressive quantization.

$260 to build it. *$0 per circuit in the long-term.*

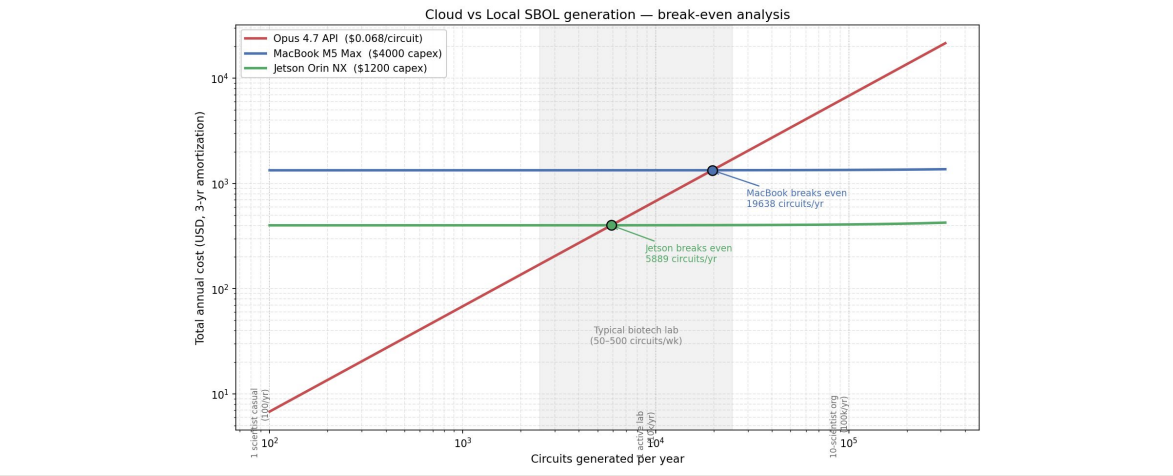
What the project cost, and where Jetson pays back vs. cloud.

TOTAL PROJECT SPEND

$260

All in. Out of pocket.

ChatGPT 5.4 (data gen)	~$253
MLX LoRA training	$0
Opus eval	~$6.80
pass	



PER-CIRCUIT COST & EFFICIENCY

PLATFORM	PER-CIRCUIT	POWER	PTS / WATT	BREAK-EVEN VS. OPUS
Opus 4.7 (API)	~$0.068	~350 W/req	0.28	—
Mac Studio Q8 LoRA	$0 (after $4k)	50 W	1.83	~19,600/yr
Jetson Orin NX + LoRA	$0 (after $1.2k)	**15 W**	**5.97**	**~5,900/yr**

Jetson is ~21× more energy-efficient than the cloud API — at 90% of the quality. At ~5,900 circuits/year (113/week), the hardware pays for itself.

Who did *what.*

Donglin Liu

BUILD & PIPELINE LEAD

Training, deployment, evaluation

Project design & scope

Dataset generation pipeline
`generate_data_llm.py` — two-stage synthesis + repair — 1,162 rows

LoRA training (both adapters)
Qwen 3.5 27B (Mac MLX) + Gemma-4 26B-A4B (Jetson) · 2,500 iters · ~7.5 h each on M5 Max

sbol_eval_v2 rubric
6-axis deterministic scorer · 100 stratified prompts

Jetson deployment
CUDA build · runtime LoRA · chat-template fix · `convert_hf_to_gguf.py` patch

`json_to_sbol3.py`
Deterministic SBOL 3.1 RDF/XML emitter

Full evaluation + 2×2 ablation (Cells A / B / C / D)

Alexander Ge

MODEL SELECTION & LOGIC DESIGN

Research, decisions, circuit-logic review

Model-family research & selection
Comparative analysis across Qwen 3.5, Gemma 4, Llama, Mistral.
Trade-off study: VRAM · quantization quality · edge throughput.
Justified **Qwen 3.5 27B** (Mac, Q8 fits in 64 GB).
Justified **Gemma-4 26B-A4B MoE** (Jetson, 4B-active fits 16 GB at Q3).

Circuit-logic design review
Topology correctness audits across the training corpus.

Behavioral-wiring validation on repressilator, toggle, QS, kill-switch, CRISPRi.

Flagged the **NAND-computes-OR** regression in kill-switch circuits.

Informed Biological-Wiring + Biological-Accuracy rubric axes.

BORROWED

Chen & Truong 2026 — prompt-only methodology
ACS Syn. Bio., DOI 10.1021/acssynbio.5c00704

SBOL 3.1 specification
SBOL community standard

Open-source stack
Qwen 3.5 · Gemma 4 · llama.cpp · MLX · pptxgenjs

AI-ASSISTED

Synthetic training data
1,162 rows via Qwen 2.5 72B + GPT-5.4
Biology judgment on Cell D outputs
Opus-as-judge rubric *(disclosed limitation, not blind)*
Debugging + codegen
Claude Code sessions, documented in session memory
This presentation
Built with Claude from Claude Code export

FIVE THINGS TO TAKE AWAY

WHY THIS IS COOL We put a frontier-capable SBOL designer on a **$1,200 fanless box** — and it ***never phones home.***

WHY IT'S CLASS-WORTHY Full train → quantize → deploy → evaluate pipeline on real SBOL3, validated across two platforms, with a reproducible deterministic rubric.

01 Edge is usable.

Jetson Gemma-4 26B-A4B + LoRA hits 90% of frontier quality at 15 watts, $0 per circuit.

02 LoRA ≈ prompt engineering — except on the tail.

At Q8 the two are ~80% redundant. At Q3 on Jetson, bare has two hard zeros; LoRA has none.

03 Quantization failure is single-token.

The Q3 bare failures were a missing comma and a `//` comment — one bad token each.

04 Valid JSON ≠ valid biology.

20-point gap between structural rubric (91.8) and expert biology judgment (71.4) on the hardest outputs.

05 Training can make logic worse.

Kill-switch topology regressed −4 points after LoRA. A 7th logic-fidelity rubric axis is the fix.

NEXT *Add the 7th (logic-fidelity) rubric axis · expand to SBOL 3.1 Sequence + Measure layers · cross-family comparison (Llama, Mistral) · wet-lab validation partner.* Total project cost to date: $260.